

# Layered Assurance for Knowledge-Graph-Mediated Newspaper Metadata: Traceable Checks, Identity Audit, and Evaluator Dependence

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## Abstract

Large language models (LLMs) can draft fine-grained metadata for digitized newspapers at collection scale, but a resolvable authority identifier can still point to the wrong entity. We study how to separate three kinds of assurance—machine checks, human identity judgment, and independent evaluation—on 203 issues of the historical newspaper *Miami Life*. An LLM extracts people, organizations, and places into a collection knowledge graph that grounds authority candidates, a deterministic gate applies lexical and temporal-plausibility checks, and a fixed generator drafts Dublin Core records. An offline replay makes all 6,401 issue–entity gate decisions traceable. Traceability is not correctness: in a blinded human audit of 406 of 917 unique candidate decisions, only 33.8% of determinate gate accepts linked the correct entity, whereas 84% of lexical and all temporal refusals were warranted. For the descriptive text, three LLM judges agree that claim precision is high but disagree on which records are better: the generator judging its own output mostly returns ties that are order reversals, while two non-generator judges prefer records drafted from OCR alone. KG-mediated metadata therefore needs separate evidence for traceability, identity validity, and descriptive quality. The results do not show that graph storage improves accuracy or that authority linking is reliable without human validation.

## Keywords

historical newspapers, knowledge graphs, metadata assurance, entity linking, human audit, large language model evaluation

## 1. Introduction

Digitization has made historical newspapers available at a scale that description has not matched. Many collections remain accessible only through issue-level records and noisy OCR: keyword search works, but typed access to the people, organizations, and places in the prose, aggregated across issues, does not. Manual description is costly, and archival practice has long accepted trade-offs between processing depth and access [1]. Large language models (LLMs) make collection-scale decomposition and drafting feasible, but fluent, structured output is not the same as a trustworthy catalog record.

Consider an authority service that returns a valid, resolvable URI for “W. E. Ellis.” The label resembles the newspaper name, yet the authority record describes a person who died decades before the issue was published. A temporal rule can expose that conflict. Harder same-name cases can satisfy both lexical and temporal checks while still denoting the wrong person. Such failures matter because a knowledge-graph edge or persistent identifier turns a local drafting error into an explicit, reusable identity assertion [2, 3, 4].

The obvious response—review every generated field—reproduces much of the work automation was meant to save. The problem is therefore not whether an LLM can produce metadata but how to distinguish three assurance questions: which checks can be executed and inspected mechanically;

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2nd Workshop on Smarter Extraction of Scholarly Metadata using Knowledge Graphs, Language Models and Agents (SESAME 2026), October 16, 2026, co-located with JCDL 2026

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whether the identity decisions are semantically correct; and whether the descriptive remainder is evaluated independently of its generator. Existing work addresses each, but the evidence types are often reported separately or collapsed into one quality score [5, 6, 7, 8].

We investigate these questions on 203 issues of *Miami Life*, a hyperlocal newspaper published from 1927 to 1949. In the workflow, an LLM decomposes each issue’s OCR into typed, issue-anchored people, organizations, and places, aggregated into a collection knowledge graph (KG); the KG grounds candidate authority edges, and a fixed LLM drafts Dublin Core records under OCR-only, authority-and-standards, and peer-experience conditions; a deterministic identity gate, separate from generation, screens the authority edges. Although the corpus is a newspaper, the same questions arise wherever scholarly-metadata pipelines attach persistent identifiers to LLM-extracted entities.

This paper makes three contributions, one per assurance question:

1. **Traceable checks (E1).** An offline replay turns the authority workflow into a 6,401-row decision trace that records why each candidate was accepted or refused; the collection-derived temporal check catches 14 wrong candidates that the lexical check passes.
2. **Measured identity validity (E2).** A blinded human audit of gate decisions shows that a fully traceable gate is not a reliable one: only 33.8% of determinate accepts link the correct entity, whereas its refusals are largely warranted.
3. **Evaluator dependence (E3).** On the same 203 issues, the generator judging its own records mostly returns order-reversal ties, whereas two non-generator judges prefer the OCR-only records; adding peer cataloging experience yields no measurable gain for any judge.

The contribution is the combined evaluation of these assurance layers on one frozen KG-mediated drafting workflow; it is not a new similarity function, and it does not establish a causal advantage of graph storage over an equivalent table. Its central finding: for identities and prose alike, the cheapest assurance—a complete trace of the identity decisions, or the generator judging its own prose—most readily reads as validation, and neither survives an independent test.

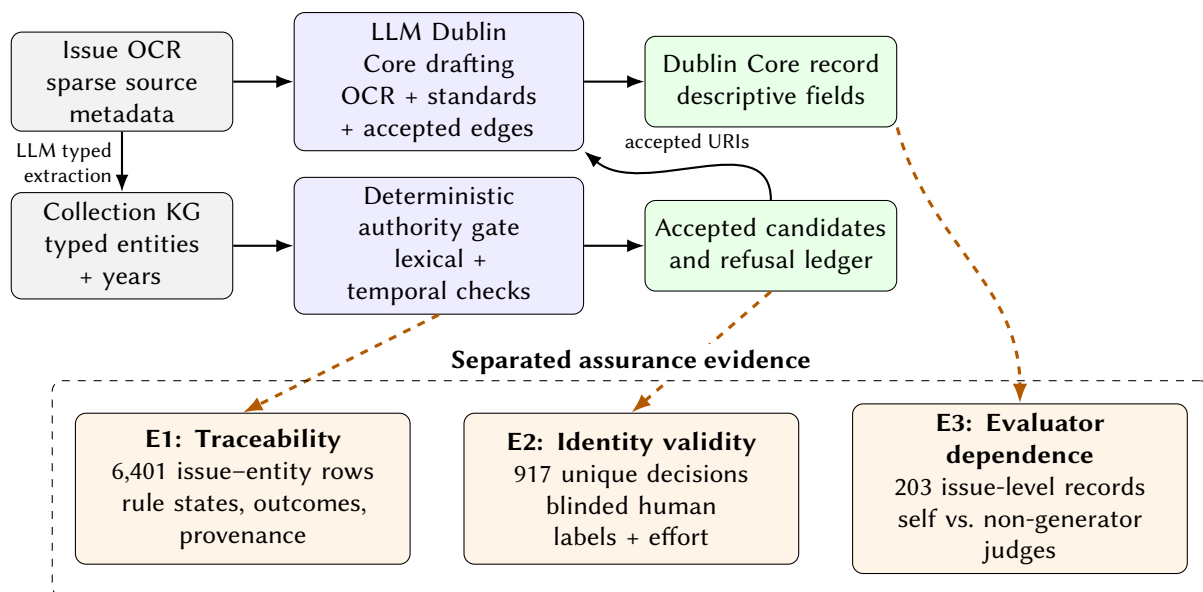
## 2. Related Work

Prior work addresses the three assurance questions separately; we review each in turn.

**LLM metadata drafting and verification burden.** LLMs can draft standards-facing library and archival metadata, but studies continue to find unauthorized terms, factual errors, and fields requiring expert remediation, with named-entity and creator fields needing the most correction [2, 3, 9, 10, 11]; earlier facial-recognition enrichment of photograph archives showed the same pattern of time savings that still required archivist confirmation [12].

**Historical entity linking and cultural-heritage validation.** HIPE established explicit entity-linking and NIL evaluation for noisy historical documents, while MELHISSA demonstrates the value of temporal evidence for anachronism-prone person links [7, 13]. SAGE combines automatic cultural-heritage annotation with professional validation and uses measured precision to filter large-scale enrichment [5]. Publishing such assertions as knowledge-graph or linked-data edges makes them reusable across systems, raising rather than resolving the validation burden. We separate the traceability of a catalog-side plausibility gate from a blinded audit of accepted and refused identities.

**Selective review and gate placement.** Selective classification treats refusal as an outcome and evaluates risk jointly with coverage [14]. Risk-ranked entity resolution and interactive entity linking show that limited human attention can be allocated more effectively than uniform review [15, 16]. Tsaneva et al. compare automated, LLM, and simulated human validators at different points in knowledge-graph construction [6]. We use accepted precision and refusal correctness as evaluation measures; routing policy is outside the scope of this study.



**Figure 1:** The production workflow and the three assurance layers. An LLM decomposes issue OCR into a collection KG whose entities the deterministic gate screens against authority candidates; only accepted edges reach drafting (C-standard shown). Gate traceability, human identity validity, and independent descriptive evaluation use different evidence and cannot be collapsed into one quality score.

**Evaluator dependence.** Pairwise LLM evaluation and atomic-claim decomposition make open-ended generation measurable [17, 18], but evaluators can favor outputs from their own model family [8, 19]. Recent digital-library work treats judge disagreement as evidence rather than noise [20]. We apply that principle to metadata descriptions by keeping the generator fixed and comparing its self-evaluation with two non-generator judges on identical items. None of these lines evaluates traceability, identity validity, and evaluator dependence on the same workflow; this study does.

### 3. Study Setting and Assurance Workflow

The workflow has three stages (Figure 1): an LLM extracts each issue’s entities into a collection KG, a deterministic gate screens their authority candidates, and a fixed LLM drafts records under three conditions.

#### 3.1. Corpus and generated records

*Miami Life* is a hyperlocal newspaper published from 1927 to 1949, digitized by FIU Libraries and the Wolfsonian Public Humanities Lab, and previously used for LLM-based discovery experiments [21]. We analyze all 203 issues available in FIU Digital Collections; the supplement lists their item identifiers and dates, from which the page images, OCR, and MODS records can be retrieved. The inputs are page-level OCR and sparse issue-level records; librarian-created MODS records provide dates, titles, and creators but not gold descriptions or authority identities. Each generated record follows the fifteen-element Dublin Core schema [22].

#### 3.2. Collection knowledge graph and authority gate

Before any record is drafted, the collection’s existing production extractor, also `gpt-5.2`, reads each issue’s OCR and emits typed, issue-anchored entities—people, organizations, and places—with roles, context snippets, and issue topics. Aggregated across the 203 issues, these form a collection knowledge graph (KG) of 8,413 entity nodes (4,939 people, 2,467 organizations, 1,007 places); 1,909 recur in more than one issue. Each node carries issue memberships, mention counts, observed activity years, and role

and context snippets, and gate-accepted authority records attach as a further node layer. Nodes are normalized typed entity surfaces (case-folded, punctuation-stripped), not resolved real-world identities: recurring surfaces aggregate into one node without coreference, and identity is asserted only by the authority gate. The KG supplements, and never replaces, the issue OCR: an issue’s own extracted entities feed authority lookup, and the collection-level aggregate supplies the gate’s activity windows.

For each of an issue’s KG entities, and separately for its extracted issue topics (which are not KG nodes), the workflow queries cached Library of Congress and GeoNames responses for authority candidates, routed by entity type (LCNAF names, LCSH topics, GeoNames places); only the top-scoring lexical candidate reaches the gate. A binary gate either permits a candidate URI to be supplied to record generation or records a refusal. Its lexical score is a hand-written combination of token coverage, Jaccard overlap, and SequenceMatcher, with manually set thresholds by entity type. For non-exact person matches, a second rule compares a candidate lifespan parsed from the authority label with the entity’s observed activity window in the collection. The gate is not an embedding model, an LLM verifier, or a calibrated probability estimator: it reads KG entity fields (surface, type, activity years) without traversing the graph. No result here isolates graph storage from an equivalent table.

### 3.3. Drafting conditions

The generator is fixed to `gpt-5.2`, as are the OCR input, output schema, and assembly code. Three conditions vary only the auxiliary knowledge supplied alongside the same OCR: `C-simple` uses issue OCR alone; `C-standard` adds a cataloging-standards block and the gate-accepted authority edges for the issue’s extracted entities; and `C-precedent` additionally adds peer cataloging episodes. The episodes are retrieved by embedding similarity, as in resolved-case retrieval [23], from a frozen store derived from publicly available external cataloging sources. The conditions predate this study; the frozen records precede a later graph-scoped retrieval refinement, which is therefore not evaluated here, and the peer-experience comparison is a scoped secondary result.

LLMs thus act only at the two ends, extraction and drafting; the identity decisions between them are deterministic and are evaluated by E1 and E2, the drafts by E3.

## 4. Evaluation

Each assurance question has its own unit and evidence (Figure 1): E1 replays 6,401 issue–entity rows (1,605 of them issue-topic rows) offline, E2 audits 917 unique candidate decisions, and E3 judges 203 issue-level records; these units are not interchangeable denominators. The frozen artifacts, per-decision audit labels, analysis notebooks, and code are available at <https://github.com/Keven1894/KG-Metadata-Generation>.

### 4.1. E1: structured-check trace

E1 asks whether every gate decision can be reconstructed. We replayed the frozen KG entity registry and authority cache through the deployed lexical and temporal rules. The replay was offline (no LLM or authority-service calls). For each selected issue–entity pair it records entity type, local years, candidate URI, score and threshold, lexical state, temporal state, final outcome, and whether an accepted URI occurs in the existing `C-standard` record; hashes identifying the run are in the supplement.

### 4.2. E2: blinded identity audit

E2 asks whether the gate’s accepts and refusals denote the right entity. The trace collapses to 917 unique (entity type, lowercased surface, candidate URI) decisions with an accept, score-refuse, or temporal-refuse outcome, randomly ordered by a fixed seed. Reviewers see the surface, authority URI and label, collection context, and OCR/PDF evidence for associated issues. They do not see the machine outcome, score, threshold, check states, refusal reason, or another reviewer’s label.

Labels are *correct*, *wrong*, *true NIL*, and *indeterminate*. True NIL means that no suitable record was found in the consulted authority sources during review; reviewers could open the candidate record and browse those sources for alternatives but were not required to search them exhaustively. Candidates labeled wrong receive a severity: *critical* if a reader would take the link as fact about a different entity, *major* if the record is related but not the entity, and *minor* if it is a harmless variant. Wrong and true-NIL judgments can also carry a residual-error category from a fixed list (supplement), required for true NIL. Five reviewers—two of the authors, the group’s program coordinator, and two student research assistants in the authors’ group—received the same written rubric and worked non-overlapping blocks beginning at positions 1, 101, 201, 301, and 401 of the seeded order. Each was asked for a single session of about two hours rather than a target count. The interface enforced sequential labeling without skipping for four reviewers; the fifth could skip but left no gaps. The reporting set is each reviewer’s longest contiguous within-block prefix.

After unblinding, accepted precision is the proportion of determinate accepted candidates judged correct, with the unique decision as unit: a decision reused across many issues counts once. Because reuse is not independent of correctness, we also report an occurrence-weighted precision over record-URI occurrences; the two answer different questions. Refusal correctness is the proportion of determinate refused candidates whose refusal was warranted. Because item time can depend on difficulty, the union of time-limited prefixes is not treated as an exact simple random sample, and the disjoint blocks are not double coding.

**Ethics.** The audit was performed by research-team members acting as research personnel; no participants were recruited, no reviewer demographic or private information was collected, and reviewer-linked labels and timing were de-identified. We treated it as annotation of system output rather than human-subjects research and did not seek IRB review.

### 4.3. E3: evaluator dependence

E3 asks whether the verdict on the descriptive fields depends on who judges them. One fixed decomposition by the generator model produces atomic claims (11,906 for C-simple, 10,816 for C-standard, 10,728 for C-precedent) that each judge checks against issue OCR, adapting supported-fact precision from FActScore [18]; claim precision is macro-averaged over issues. Authority URIs are removed because they do not occur in OCR and are evaluated separately in E2. The pairwise instrument compares URI-stripped records in both orders. Following MT-Bench, order reversals count as ties [17].

The panel contains the generator self-judge (gpt-5.2), a cross-provider judge (claude-opus-4-8), and a same-provider non-generator judge (gpt-5.5); only the self-judge runs at temperature 0, because the other two providers fix or reject the parameter (sampling details are in the supplement). We report unconditional win/tie/loss counts before decided-vote percentages, Wilson intervals, exact binomial tests, and swap consistency. For 182 of 203 issues, OCR exceeded 60,000 characters, so all conditions were judged over the same head-and-tail OCR window.

## 5. Results

### 5.1. E1: Traceable checks

The replay gives a complete, hash-identified account of the gate (Table 1). Temporal and lexical evidence are not redundant: 14 temporal failures pass the lexical threshold. Replay accepts cover all 2,837 authority-URI occurrences in the existing C-standard records.

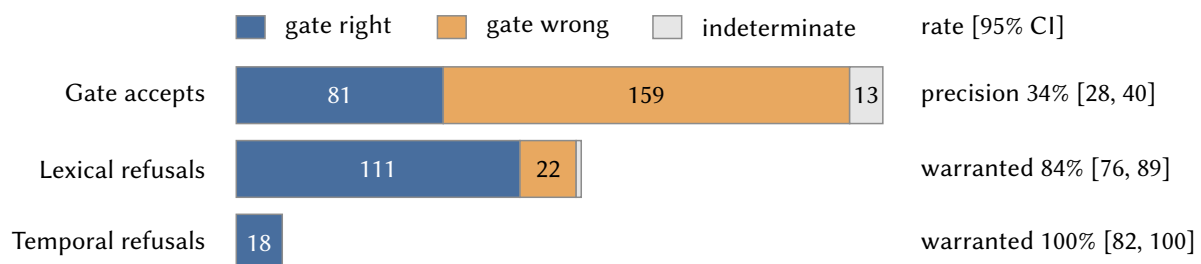
### 5.2. E2: Identity validity

The blinded audit shows a strongly asymmetric gate: most of its accepts do not denote the right entity, while most of its refusals are warranted (Figure 2).

**Table 1**

E1 offline replay over the frozen 203-issue corpus. Outcomes are issue–entity (including issue-topic) rows, not unique identities.

| Trace outcome or downstream check         | Count         |
|-------------------------------------------|---------------|
| Accepted candidate                        | 2,948         |
| Lexical-score refusal                     | 827           |
| Temporal refusal                          | 55            |
| No candidate                              | 2,571         |
| All trace rows                            | 6,401         |
| Temporal failures otherwise passing score | 14            |
| Existing record-URI occurrences covered   | 2,837 / 2,837 |



**Figure 2:** Blinded identity audit, unblinded against the gate: 406 unique decisions at the analysis freeze. Segments are counts, colored by whether the deterministic gate’s decision was right. On accepted rows a wrong gate means the link denotes the wrong entity (130) or an entity with no suitable record (29); on refused rows it means the blocked URI was in fact correct. Rates exclude indeterminate labels (the thin sliver on the lexical row) and carry Wilson intervals.

The audit frame contains 917 unique candidate decisions: 561 accepts, 319 lexical refusals, and 37 temporal refusals, across 451 person, 237 organization, 210 place, and 19 subject/topic decisions. At the analysis freeze, five reviewers had labeled 406 decisions (44.3%; 253 accepts, 135 lexical and 18 temporal refusals) as five contiguous within-block prefixes with no skipped item. Review took 1.37 minutes per decision (1.27–1.48 across reviewers; wall-clock gaps between submissions, excluding pauses over five minutes), so the full 917-decision frame projects to about 21 active review hours.

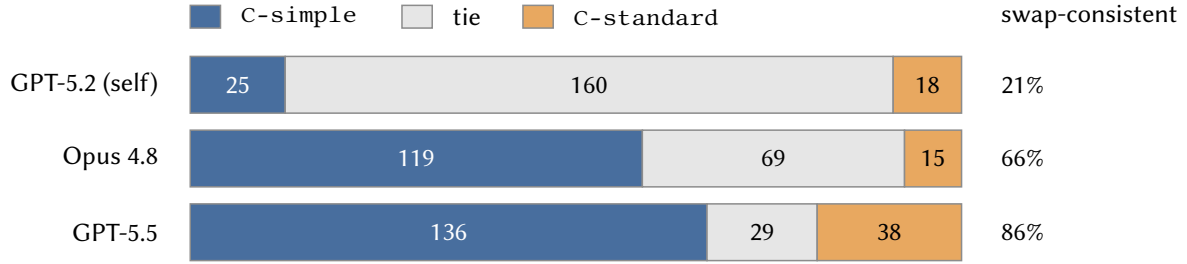
The labeled set matches the population on outcome mix (62% vs. 61% accepts) and entity type, and labeled accepts match unlabeled ones in hidden score (Kolmogorov–Smirnov  $D = 0.036$ ); stopping tied to unobserved difficulty cannot be ruled out.

Of 240 determinate accepted decisions, 81 denote the right entity: **accepted-link precision is 33.8%** (Wilson 95% CI [28.1%, 39.9%]) per unique decision. The other 159 are 130 wrong identities and 29 entities with no suitable record in the consulted sources; 13 are indeterminate.

Occurrence weighting answers a different question. Weighting each decision by its record-URI occurrences gives 58.6% (865 of 1,476 occurrences attached to the reviewed decisions). It is higher because a few correct decisions are reused very often (mean 10.7 vs. 4.3); it is not corpus-wide record precision.

Refusals fare differently: 111 of 133 lexical refusals were warranted (83.5%, [76.2%, 88.8%]), as were all 18 temporal refusals (100%, [82.4%, 100%]). The 22 false lexical refusals blocked a correct URI.

The accepted-link errors are often consequential. Of the 130 accepted candidates judged wrong, 111 (85.4%) are critical and none is minor; across accepted and refused candidates, 209 of 242 wrong labels are critical. Among the 141 wrong-or-NIL candidate labels carrying a recorded mismatch reason, across accepted and refused candidates alike, same-name same-era collisions (59, 41.8%) and entities with no authority record (49, 34.8%) dominate; organization alias or parent confusion accounts for 14. Reviewers diverged only in true-NIL versus wrong calls (0% to 38%), which moves neither headline rate. Per-reviewer accepted precision spans 25.0–41.3% with overlapping intervals, leaving any one reviewer



**Figure 3:** Unconditional pairwise outcomes for C-simple versus C-standard over all 203 issues, judged in both presentation orders (segment labels are issue counts). A tie is almost always a reversal between the two orders; swap-consistent is how often a judge gave the same verdict twice.

**Table 2**

E3 pairwise statistics by judge (203 issues): Figure 3 conditioned on decided votes (Wilson CI, exact binomial test).

| Judge          | C-simple vs. C-standard, decided votes |        |                       |            |
|----------------|----------------------------------------|--------|-----------------------|------------|
|                | C-simple wins                          | 95% CI | exact $p$             | swap-cons. |
| GPT-5.2 (self) | 25/43 (58%)                            | 43–72% | 0.36                  | 21%        |
| Opus 4.8       | 119/134 (89%)                          | 82–93% | $2.9 \times 10^{-21}$ | 66%        |
| GPT-5.5        | 136/174 (78%)                          | 71–84% | $3.9 \times 10^{-14}$ | 86%        |

out keeps it within 31.1–35.7%, and counting every indeterminate as correct raises it only to 37.2%. The asymmetry therefore does not hinge on any one reviewer or on the indeterminate calls.

### 5.3. E3: Evaluator dependence

All three judges assign high OCR-supported atomic-claim precision across the generation conditions (0.948–0.991), yet their pairwise conclusions diverge: on C-simple versus C-standard, the self-judge returns 160 ties, against 69 and 29 for Opus and GPT-5.5 (Figure 3).

On decided votes (Table 2), both non-generator judges clearly favor C-simple, while the self-judge’s 25 of 43 is indistinguishable from chance; at 21% swap consistency, its ties are mostly order reversals rather than judgments of equal quality. The peer-experience comparison is a scoped negative result. On C-standard versus C-precedent, decided votes split 12–10 (self), 56–53 (Opus), and 65–65 (GPT-5.5), with 181, 94, and 73 ties; C-simple versus C-precedent mirrors the primary comparison (counts in the supplement).

## 6. Discussion and Limitations

**Traceability is not validation.** The collection KG makes every entity and authority assertion explicit and reusable, yet a complete trace does not make a lexical heuristic a probability of correctness. The rules’ inputs, thresholds, and refusal reasons can be examined, and the 14 incremental temporal catches show that collection chronology detects failures lexical matching misses. The same transparency sets the claim boundary. Same-name, same-era candidates cannot be separated from correct matches by thresholding these features: the surfaces match and the lifespans overlap. Deciding such a case needs evidence the gate never reads (the surrounding article text). Such collisions are the largest reason among wrong-or-NIL accepts (37 of 81 with a recorded reason). Human audit is thus an independent measurement layer, not evidence that the gate “verified” an identity. At this precision, the gate should not publish links without human validation; its defensible role is candidate screening, refusal, and provenance capture.

**Evaluator choice changes the reported finding.** The self-judge alone would report “no difference”; both non-generator judges prefer the OCR-only description. This does not establish user preference—judges may reward denser prose catalogers would not—but generator self-evaluation cannot certify “no difference.” As C-standard drafted with the edges E2 audits, link errors may have entered its prose; the URI-stripped view keeps the comparison independent of the audit; propagation is left open.

**Implications for digital-library practice.** We recommend three separations: retain the evidence and reason for every gate decision rather than treating a resolver hit as truth; audit accepted as well as refused candidates; and use independent evaluators for descriptive claims, reserving identity for human judgment. The same separation applies wherever a pipeline attaches persistent identifiers to extracted entities: library cataloging workflows that add LCNAF or GeoNames links to digitized collections, and scholarly knowledge graphs that link extracted authors, institutions, and works to ORCID, ROR, or DOI.

**Limitations.** One generator and one hyperlocal collection were studied, so the rates do not transfer to other models or collections; what transfers to other newspapers is the protocol—its three layers, units, and audit design—not the measured values. The gate’s lexical score and thresholds are hand-written and uncalibrated, and because only the top lexical candidate reaches the gate, E2 cannot separate retrieval and ranking failures from gate failures. Because KG nodes aggregate by normalized surface, an activity window may merge homonymous mentions, which can only mask temporal conflicts. E2 evaluates linking given the extracted surfaces, not the extraction itself. Its blocks are disjoint, so inter-reviewer agreement cannot be computed; the sensitivity diagnostics bound the risk from time-limited stopping but do not replace double coding, and audit throughput is not net labor savings against manual cataloging. Because C-standard varies the standards block and the accepted edges together, the judges’ preference cannot be attributed to either. Finally, the absence of a peer-experience benefit does not distinguish insufficient episode coverage, retrieval design, and true lack of transfer.

## 7. Conclusion

A collection KG makes entity and authority assertions explicit and reusable, but fluent output and resolvable identifiers do not establish identity correctness. Separating the layers shows why: collection structure supports traceable lexical and temporal checks; blinded review finds that the fully traceable gate admits more wrong links than right ones while its refusals are largely warranted; and non-generator judges overturn the generator’s verdict of “no difference,” preferring the OCR-only prose. Libraries adopting LLM metadata should keep these evidence types separate and validate accepted links.

## Acknowledgments

FIU Libraries funded the model API usage. We thank Hong Chen and Chunhan Cheng, student research assistants, and Taylor Bonachea, program coordinator, for their work on the identity audit.

## Declaration on Generative AI

During the preparation of this work, the authors used GPT-5.5 and GPT-5.6 Sol (OpenAI), Claude Fable 5.1 and Claude Opus 5.5 (Anthropic), and Grok 4.7 (SpaceXAI) in order to: Drafting content, Improve writing style, Paraphrase and reword, Grammar and spelling check, and Peer review simulation. The authors rewrote the drafted passages and checked every reported number against the frozen analysis artifacts in the supplementary material. The figures were drawn by the authors in  $\text{\LaTeX}$ /TikZ. Separately, gpt-5.2, claude-opus-4-8, and gpt-5.5 are components of the studied workflow and its evaluation (Sections 3 and 4); they are research instruments, not writing aids. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication’s content.

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